

MUHAMMAD UZAIR

Python Full Stack Developer - Django & FastAPI · REST APIs · AI / LLM Integrations

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PROFESSIONAL SUMMARY

Python Full Stack Developer with 3+ years of experience designing, building, and deploying production web applications, RESTful APIs, and AI-powered services using Python, Django, Django REST Framework, FastAPI, React, and PostgreSQL. At AI-Nafi, built backend services for the AI-Baseer internal analytics platform: payment reconciliation across Stripe, UBL IPG, and Easypaisa, Thinkific LMS webhooks, JWT authentication, Redis caching, and automated report exports to AWS S3. Delivered 10+ client projects end to end (pharma CRM, payroll, e-commerce, CMS, and coupon platforms), from database design to Docker and Gunicorn deployment. Hands-on with LLM integrations (Google Gemini, OpenAI GPT, RAG with Qdrant), OCR and computer vision (Tesseract, OpenCV), web scraping, and data pipelines with Pandas.

TECHNICAL SKILLS

Languages: Python, JavaScript, TypeScript, SQL, HTML5, CSS3

Backend & APIs: Django, Django REST Framework (DRF), FastAPI, GraphQL (Strawberry), WebSockets, RESTful API design, JWT authentication (SimpleJWT), OpenAPI/Swagger (drf-spectacular), webhooks, Pydantic

Frontend: React, Next.js, Vue.js, Tailwind CSS, Docusaurus, responsive UI, vanilla JavaScript

Databases & Caching: PostgreSQL, MySQL, SQLite, Redis, SQLAlchemy ORM, Django ORM, Qdrant (vector database), query optimization

AI / ML & Data: Google Gemini API, OpenAI GPT API, LLM agents, Retrieval-Augmented Generation (RAG), prompt engineering, Tesseract OCR, OpenCV, Hugging Face Transformers, Pandas, NumPy, Streamlit

Cloud & DevOps: AWS (S3, boto3, django-storages), Docker, Docker Compose, Gunicorn, WhiteNoise, Heroku, Linux, Git, GitHub, Sentry monitoring

Integrations & Automation: Stripe, Easypaisa, UBL IPG, Thinkific, Open edX (OLX/XML), Selenium, BeautifulSoup, Requests, WeasyPrint (PDF), CSV/XLSX import-export

Testing & Practices: pytest, pytest-django, unit testing, flake8, black, code reviews, Agile/Scrum, technical documentation

PROFESSIONAL EXPERIENCE

Full Stack Developer (Python / Django) | AI-Nafi

2023 - Present · Karachi

- Built and maintained RESTful APIs with Django REST Framework for **AI-Baseer**, AI-Nafi's internal analytics platform, serving users, enrollments, payments, products, and trainers data for AI-Nafi and Islamic Academy.
- Developed payment search, validation, and reconciliation services across **Stripe, UBL IPG, and Easypaisa** gateways, including renewal tracking and date-range reporting used by finance and sales teams.
- Integrated **Thinkific webhooks** (user and enrollment events) to sync LMS data in real time, eliminating manual data entry between systems.
- Implemented JWT authentication with secure cookie-based tokens and group-based permission classes to enforce role-based access control (RBAC) across internal APIs.
- Improved API response times with **Redis caching** and query optimization on MySQL/PostgreSQL; tracked production errors with **Sentry**.
- Automated CSV/XLSX report generation with Pandas and uploaded exports to **AWS S3** via boto3; added admin import/export and date-range filters for operations staff.
- Built a Python automation tool that converts CSV course outlines into **Open edX OLX XML** (course, chapter, sequential, vertical) with UUID-based files, turning hours of manual course setup into a single command.
- Wrote **Selenium** automated test scripts for payment flows (debit card, direct deposit, Easypaisa, ePay) to catch checkout regressions before release.

Freelance Python Full Stack Developer | Self-Employed

2024 - Present · Remote

- Delivered 10+ production web applications for clients in pharma, outdoor advertising, retail, e-commerce, and digital media using Django, Tailwind CSS, PostgreSQL, Gunicorn, and WhiteNoise.
- Built **Human 4 Health Pharma CRM** with 8 Django apps (doctors and visits, medical representatives, distributors, medical stores, products and batches, sales, HR, analytics), including expiry alerts, performance snapshots, and a CSV import tool with preview and validation.

- Developed a multi-company **Payroll Management System** with Pakistan income-tax slabs, provident fund, gratuity, advance deductions, bulk payroll processing, and PDF payslips (WeasyPrint); containerized with Docker on PostgreSQL.
- Built an e-commerce REST backend (DRF, SimpleJWT) with products, collections, coupons, orders, and reviews, storing media on **AWS S3** via django-storages.
- Shipped SEO-focused coupon and blog platforms with real-time community voting, search and filters, CMS-managed content, newsletters, and admin import/export (CSV, XLSX, JSON).

CUSTOM PROJECTS

AI Flyer Offer Extraction Agent - LLM / Computer Vision Private repository

- Built an AI agent that sends retail flyer images to **Google Gemini** (multimodal) and returns structured JSON of offers, prices, and validity dates through a FastAPI service with async job management and Pydantic validation.
- Built a companion **Flyer Offer Detection API** (Django REST Framework, OpenCV, Tesseract) that detects product boxes, runs multi-language OCR (Arabic, English, Urdu, Hindi, French, Spanish) with per-offer confidence scores, and stores cropped offer images; benchmarked **DeepSeek-OCR** (Hugging Face Transformers) as an alternative.

Technologies: Python, FastAPI, Google Gemini API, Django REST Framework, OpenCV, Tesseract OCR, Transformers, Pillow, Pydantic

AI-Driven E-Book Platform with RAG Chatbot - Hackathon github.com/Uzairrrrrr/hackathon-1

- Built an interactive e-book with a **RAG chatbot**: book content indexed in Qdrant and answered by Gemini, with JWT signup/login, user profiling, AI-personalized content, and cached Urdu translation.

Technologies: FastAPI, Google Gemini, Qdrant, SQLAlchemy, JWT, React, TypeScript, Docusaurus, Docker

TaskFlow - Real-Time Task Management App github.com/Uzairrrrrr/taskflow

- Built a full-stack task manager exposing both REST and **GraphQL** APIs, with **WebSocket** live updates, JWT authentication, filtering by status, priority, and category, and a statistics dashboard.

Technologies: FastAPI, Strawberry GraphQL, WebSockets, SQLAlchemy, Pydantic, React, Tailwind CSS

AI Job Description Parser github.com/Uzairrrrrr/ai-job-parser

- Built an LLM pipeline that reads job listings from CSV and uses the **OpenAI GPT API** to extract job summaries, key responsibilities, and required skills into a clean, analysis-ready dataset.

Technologies: Python, OpenAI API, Pandas, Pydantic, tqdm

Academic Paper Data Scraper github.com/Uzairrrrrr/released-papers-data-scraper

- Built a configurable data collection pipeline on the **OpenAlex API** that gathers 1,000+ papers (2023-2025) with citations, references, and related works, de-duplicates records, and exports CSV and JSON.

Technologies: Python, Requests, Pandas, REST APIs, JSON

OTHER PROJECTS

- **DRF E-commerce API:** nested categories (django-mptt), OpenAPI docs (drf-spectacular), and tests with pytest and pytest-django.
- **Recipe API and Django + Vue app:** containerized with Docker and Docker Compose, backed by MySQL, with unit tests.
- **Streamlit apps:** personal library manager (SQLite, Pandas, live stats dashboard) and a game hub; built during the GIAIC program.

EDUCATION

Bachelor of Computer Science (Graduated) - Federal Urdu University of Arts, Sciences & Technology, Karachi, Pakistan Mar 2020 - Mar 2024

Intermediate, Pre-Engineering - DJ Sindh Government Science College, Karachi, Pakistan Aug 2017 - Jul 2019